BANKING CUSTOMER & RISK ANALYTICS

Business Intelligence, Portfolio Insights & Default-Risk Modeling

DETAILED BUSINESS REPORT

Data scope: 5 datasets • approximately 690,100 source records
Methods: Data validation • EDA • classification modeling • business recommendations

Executive Summary

This report presents an end-to-end Banking Customer & Risk Analytics project combining data quality checks, exploratory data analysis (EDA), SQL-oriented analytics, dashboard-ready business metrics, and an initial machine-learning experiment to identify loans associated with default risk. The project covers five datasets: 75,000 customers, 85,000 accounts, 30,000 loans, 500,000 transactions, and 100 branches—approximately 690,000 records in total.

The EDA highlights differences across account types, concentration in payment channels, high-value credit transactions, and a loan portfolio in which 452 loans are labelled "Defaulted" in the loan-status field. A separate target field, default_flag, labels 1,617 of 30,000 loans as defaults (5.39%). These two measures do not agree and must be reconciled before any official business reporting or operational use.

For a preliminary default-prediction experiment, Logistic Regression and Random Forest were trained using a stratified 80/20 split. Logistic Regression achieved ROC-AUC 0.7777 and recall 72.45% at a 0.50 threshold; Random Forest achieved ROC-AUC 0.7648 and recall 6.81% at the same threshold. The results suggest Logistic Regression is the stronger initial candidate for further validation, but the model is not production-ready.

Recommended priorities are to validate target definitions and monetary units, review credit-risk drivers with appropriate controls, monitor high-value payment reliability, and use the model only as decision support after independent validation, calibration, fairness checks, and business-cost assessment.

1. Business Context and Objectives

Banks need reliable visibility into customers, accounts, lending exposure, branch operations, and payment activity. A joined analytical view can help stakeholders monitor portfolio quality, identify areas for investigation, and improve operational decisions.

- Build a repeatable data preparation and validation workflow across five banking datasets.
- Summarize customer, account, loan, branch, and transaction performance using clear metrics.
- Explore factors associated with the available loan default label.
- Compare baseline classification models and understand the trade-off between missed defaults and false alarms.
- Translate analysis into practical, measurable follow-up actions.

Scope note: the data is treated as project/synthetic analytical data unless its provenance is independently documented. Findings should not be presented as verified facts about a real bank.

2. Data Scope and Quality

Dataset	Rows	Analytical use
Customers	75,000	Customer profile and segmentation
Accounts	85,000	Account types and balances
Loans	30,000	Loan portfolio and default labels
Transactions	500,000	Payment volume, value, status and channel
Branches	100	Geographic footprint and staffing

The combined source footprint is approximately 690,100 rows. The project validation checks reported no missing values and no duplicate primary keys in the reviewed source tables, and no orphan-key issues were reported in the QA summary. These checks are useful but do not by themselves establish that all values are semantically valid.

Important data-quality caveats:

(1) loan_status contains 452 "Defaulted" loans, while default_flag contains 1,617 "Yes" values; definitions, timing, and derivation must be investigated.

(2) The account-balance exploration used dollar-style labels, while transaction values were described in rupees; confirm currency and units before comparing or publishing monetary totals.

(3) The account data includes 16,903 negative balances, especially loan accounts and credit-card accounts; negative values may be valid for some account types but require business-rule validation.

(4) The reported transaction totals and payment-status counts should be rechecked against source aggregation queries before final publication.

3. Exploratory Data Analysis Findings

3.1 Account balances and product mix

Account balances were concentrated broadly between 0 and 50,000, with the largest concentration around 10,000–30,000. The explored range extended approximately from −200,000 to +250,000. Average balances varied materially by account type: Fixed Deposit about 125,000; Savings about 38,000; Checking about 18,000; Credit Card about −7,513; and Loan Account about −100,535.

Business interpretation: account types have different balance semantics. A negative loan balance may represent outstanding debt rather than a data error, while negative credit-card balances may represent amounts owed or other accounting conventions. Segmenting by account type is essential; do not interpret all negative balances as customer distress without validating the data dictionary.

3.2 Branch footprint and staffing

The branch footprint was reported as South 26 branches, North 22, and Central 12. Maharashtra had 13 branches; several other states were reported with seven branches each. Account volumes across branches were relatively close, approximately 3,370–3,449 accounts per branch. Staffing varied from about 32 at Bhopal to 167 at Aurangabad.

Business interpretation: staffing differences may reflect branch size, role mix, service model, or reporting issues. Before reallocating employees, compare workload indicators such as transactions per employee, customer visits, service times, open cases, and branch operating hours. The available summary alone does not prove overstaffing or understaffing.

3.3 Loan portfolio and credit risk

The portfolio contains 30,000 loans. Reported loan-status counts are approximately 20,948 Active, 8,600 Closed, and 452 Defaulted. Home loans are the largest product group at about 10,600, followed by Personal loans at 7,400 and Car loans at 6,100; Business and Education loans are around 3,000 each.

Income-tier default rates were reported in a range of approximately 4.23%–5.61%, with the ₹10 lakh–₹20 lakh tier containing 15,289 loans and a reported rate of 5.61%. This pattern does not establish that income alone drives default. Credit score, debt burden, interest rate, loan size, repayment history, loan type, and data timing should be evaluated together.

3.4 Transactions and payment operations

The transaction dataset contains 500,000 records with a reported total value of ₹6.02 billion. Credit transactions were described as contributing approximately 83% of transaction value, with average transaction value around ₹50,490.01; debit transactions represented approximately 80% of transaction volume, averaging around ₹2,505.35. UPI was the most common channel at roughly 250,000 transactions, followed by Debit Card at about 100,000 and Credit Card at about 75,000.

Approximately 460,053 transactions were successful, 24,930 failed, and 15,017 pending. This implies a failure rate of about 4.99% when failed transactions are divided by all 500,000 transactions. The exact rate depends on the

denominator and whether pending transactions are included. High-value credit payments and frequently used UPI journeys merit focused reliability monitoring.

4. Machine Learning: Default-Risk Classification

4.1 Objective and target

The model experiment predicts the available default_flag label for each loan. The target distribution is 28,383 "No Default" and 1,617 "Default" records, a positive class rate of 5.39%. This is an imbalanced classification problem: a model can obtain high accuracy while still missing most defaults.

Ten features were selected: loan_type, loan_amount, interest_rate, emi, credit_score, age, annual_income, occupation, marital_status, and customer_segment. Identifiers and the fields loan_status and remaining_balance were excluded from the feature set to reduce direct leakage risk. However, the feature timing and business meaning still need review to ensure every feature would be available at the intended prediction point.

4.2 Method

- Stratified train/test split: 80% training (24,000 rows) and 20% testing (6,000 rows).
- Preprocessing: scale numerical features and one-hot encode categorical features.
- Models: class-weight-balanced Logistic Regression and Random Forest (200 trees).
- Evaluation: accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.
- Threshold analysis: compare recall and false-alarm burden at multiple probability cut-offs.

4.3 Model comparison at a 0.50 threshold

Metric	Logistic Regression	Random Forest
Accuracy	70.23%	93.57%
Precision	12.12%	20.56%
Recall	72.45%	6.81%
F1-score	20.76%	10.23%
ROC-AUC	0.7777	0.7648
True positives	234	22
False negatives	89	301
False positives	1,697	85
True negatives	3,980	5,592

Interpretation: Logistic Regression identifies more of the labelled defaults at the 0.50 threshold and has a slightly higher ROC-AUC, but it flags many non-default loans for review. Random Forest produces fewer false positives but misses most labelled defaults at this threshold. For risk screening, accuracy alone is not a sufficient selection metric.

4.4 Logistic Regression threshold trade-off

Threshold	Precision	Recall	False alarms	Missed defaults
0.20	7.36%	96.28%	3,914	12
0.30	8.87%	91.64%	3,041	27
0.40	10.64%	85.76%	2,326	46
0.50	12.12%	72.45%	1,697	89
0.60	13.24%	55.11%	1,166	145
0.70	15.59%	40.25%	704	193

Lower thresholds catch more labelled defaults but create more false alarms and review workload. Higher thresholds reduce false alarms but miss more defaults. The threshold must be selected using a validation set and a documented

cost model—not by choosing a threshold based on the test set. The table is exploratory and should be reproduced with validation-only threshold selection before reporting performance as final.

4.5 Feature signals

The strongest Random Forest feature importances were credit_score (0.3177), interest_rate (0.1544), EMI (0.1078), annual_income (0.0984), age (0.0874), and loan_amount (0.0835). In Logistic Regression, credit_score had the largest-magnitude coefficient (−1.2782), consistent with higher scores being associated with lower predicted default probability in this fitted model, subject to feature scaling and encoding.

These are model associations, not causal effects. Importance values can be affected by correlated predictors, target construction, sampling, preprocessing, and data quality. Occupation, marital status, and customer segment should be handled carefully and assessed for fairness, policy relevance, and legal permissibility before any real-world use.

5. Business Recommendations

Priority	Recommendation	Suggested measure
1. Data governance	Reconcile default_flag and loan_status definitions; document currency, units, and as-of dates.	One approved target definition; documented data dictionary; reconciliation checks passing
2. Credit risk review	Use credit score alongside debt-to-income, repayment history, loan type, EMI burden, and loan amount. Validate whether these fields are available at decision time.	Default rate and loss rate by risk band; stability over time
3. Human-in-the-loop screening	Treat model output as a review signal only; do not automatically reject or price loans from this prototype.	Recall, precision, review volume, override rate, and adverse-impact monitoring
4. Payment reliability	Monitor UPI success/failure and high-value credit transactions by channel, time, issuer/acquirer, and failure reason where available.	Failure rate, p95 processing time, value-weighted failures, recovery rate
5. Branch operations	Compare staffing with workload and service outcomes before making resource changes.	Transactions per employee, wait time, open cases, SLA attainment
6. Customer/product analytics	Investigate high-balance Fixed Deposit customers for suitable service or wealth-product engagement, subject to consent and suitability rules.	Eligible-customer engagement, conversion, complaints, and retention

Recommended order of execution: first resolve data definitions and units; then rerun the descriptive metrics; then evaluate the model on a clean, temporally appropriate split; finally pilot a human-reviewed workflow with explicit monitoring and rollback criteria.

6. Limitations and Next Steps

- Target inconsistency: 452 "Defaulted" loan_status rows versus 1,617 positive default_flag rows. The project must explain this discrepancy.
- Potential synthetic or simulated data: results may not generalize to a real banking portfolio.
- Currency ambiguity: account balance labels and transaction currency need reconciliation.
- Class imbalance: default prevalence is only 5.39%, making accuracy misleading.
- Threshold selection: threshold results were explored on the test set; final threshold selection should use a validation set or cross-validation.
- Feature timing: verify that each predictor is known before the prediction date; avoid post-outcome or post-decision leakage.

- Limited model validation: no external test set, time-based backtest, probability calibration, fairness audit, or independent replication is included yet.
- Operational costs: false positives create manual review work, while false negatives may allow risk to go undetected. The bank-specific costs are not currently known.

Next steps: (1) reconcile and document the target; (2) validate units and business rules; (3) establish a baseline and time-based holdout if dates are available; (4) tune the threshold using validation data and expected business costs; (5) inspect calibration and subgroup performance; (6) document model limitations and human oversight; (7) refresh the dashboard and report after data checks pass.

7. Conclusion

The project demonstrates an end-to-end analytical workflow across customer, account, loan, branch, and transaction data, with an initial ML comparison for default-risk screening. The most useful immediate output is not an automated credit decision; it is a repeatable data-quality process, clearer portfolio reporting, and a transparent view of the trade-off between catching potential defaults and creating false alarms.

Logistic Regression is the better candidate of the two models in this initial experiment by recall and ROC-AUC at the reported threshold, but further validation is necessary before any operational use. The project should be presented as a portfolio demonstration and decision-support prototype, not a validated production credit-risk system.

Appendix A. Key Metric Reference

Metric	Value	Note
Total records across five tables	690,100	Sum of row counts; source records may have different grains
Loan default_flag positives	1,617 / 30,000 (5.39%)	Target used in ML experiment
loan_status = Defaulted	452 / 30,000 (~1.51%)	Different field; reconcile with target
Transactions	500,000	Reported total value ₹6.02B
Successful transactions	460,053	Reported count
Failed transactions	24,930	Approximately 4.99% of all transactions
Pending transactions	15,017	Included in all-transaction denominator
Logistic Regression ROC-AUC	0.7777	Exploratory test result
Random Forest ROC-AUC	0.7648	Exploratory test result

Appendix B. Suggested GitHub Structure

Suggested report placement within the project repository:

```
reports/
  business_report/
    Banking_Customer_Risk_Analytics_Business_Report.pdf
    Banking_Customer_Risk_Analytics_Business_Report.docx
  ml/
    model_comparison.csv
    threshold_comparison.csv
    logistic_regression_coefficients.csv
    random_forest_feature_importance.csv
    threshold_comparison.png
```

Before committing, ensure the report's metrics match the latest notebook outputs, avoid committing credentials or personal data, and include the report source/refresh date in the repository README.